

## SL-Assignment-1

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1. Using for loop, write a Java program for n factorial

Program:-

```
import java.util.Scanner;

public class Factorial {
    public static void main(String []args){
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter the number: ");
        int num=sc.nextInt();

        int fact=1;
        for(int i=1;i<=num;i++){
            fact=fact*i;
        }
        System.out.println("Factorial of "+num+" is "+fact);
    }
}
```

Output:-

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Fi
Enter the number: 5
Factorial of 5 is 120

Process finished with exit code 0
|
```

2. Modify the program above using a while loop so it prints out all of the factorial values that are less than 1000.

Program:-

```
public class ModifiedFact {
    public static void main(String []args){
        int fact=1;
        int num=1;
        while(num!=0) {
            fact=1;
            int i = 1;
            while (i <= num) {
                fact = fact * i;
                i++;
            }
            if(fact<=1000) {
                System.out.println("Factorial of " + num + " is " + fact);
                num++;
            }
            else{
                break;
            }
        }
    }
}
```

Output:-

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.
Factorial of 1 is 1
Factorial of 2 is 2
Factorial of 3 is 6
Factorial of 4 is 24
Factorial of 5 is 120
Factorial of 6 is 720

Process finished with exit code 0
```